As per ohm's law $V = IR_1+IR_2+IR_3$

$V=I(R_1+R_2+R_3)$

$$\frac{V}{I} = R_1 + R_2 + R_3$$

$$R = R_1 + R_2 + R_3$$

$$R = \frac{V}{I}$$

Where R is the total resistance of the series circuit

That is total resistance R is the sum individual resistors present in the series circuit.

4. The current flows in the individual resistors will be $I = \frac{V}{R}$
5. The current cannot flow through the circuit, if there is any disconnection in any one point of the circuit.

1.5.1 Uses of Series Connection

1. Series connection is used in serial sets as decorative lamps which are used in festivals and functions.
2. It is used to find out the defect in electrical appliances such as tube light, mixie, electric stove and fans.

Problems

Two resistors of values 100 Ω and 200Ω are connected in series with a 300 volts supply. Prove that the sum of the voltage drops in the circuit is equal to the supply voltage.

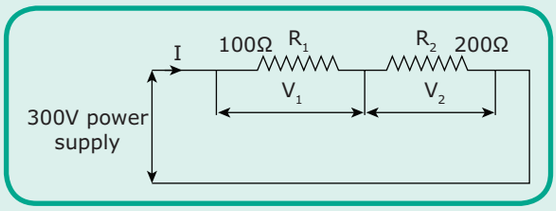


Figure 1.21

Total Resistance R = R1+R2

= 100+200

R = 300Ω

Current $I = \frac{V}{R} = \frac{300}{300} = 1A$

Voltage drop $V_1 = I \times R_1 = 1 \times 100 = 100V$

Voltage drop $V_2 = I \times R_2 = 1 \times 200 = 200V$

V = V1 + V2

300 V = 100V + 200 V

300 V = 300 V

The sum of the voltage drops in the circuit is equal to the given supply voltage. Hence, it is proved.

1.6 RESISTORS CONNECTED IN PARALLEL

When two resistors are connected one after the other as shown in the Figure 1.22 (the first ends and last ends are connected separately), the number of current flowing paths (two) is equal to the number of resistors present in the circuit.

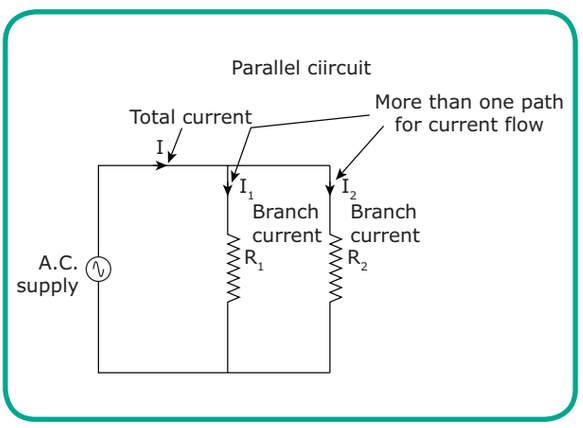


Figure 1.22

Figure 1.22 shows R_1 and R_2 are connected in parallel. When the voltage 'V' is applied in between the ends of the circuit, the following points are to be noted.

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